Lesson 6: Paper Helicopter Independent Skills Challenge

Unit

Working Like a Physicist: Collecting, Processing, and Presenting Data

Lesson focus

Students independently apply the data handling skills developed across the unit: variables, tables, repeated measurements, averages, graph choice, plotting, best-fit lines, and anomalous results.

Assumptions

Students have already practised categoric data, continuous data, bar charts, line graphs, and best-fit lines.

Practical title

Paper helicopter investigation

Recommended investigation question

How does wing length affect the fall time of a paper helicopter?

Why this practical works well

This practical is fun, cheap, and easy to run in a standard classroom or lab. It gives students a chance to design a table, collect repeated results, calculate averages, choose a graph, and plot a line graph.

Learning objectives

By the end of the lesson, students should be able to independently:

1. Identify independent, dependent, and control variables.
2. Draw a suitable results table.
3. Collect at least 3 trials for each value of the independent variable.
4. Calculate mean values.
5. Choose the correct graph type.
6. Plot a line graph accurately.
7. Draw a best-fit line.
8. Identify possible anomalous results.
9. Write a conclusion based on the data.

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Key vocabulary

Keyword	Student-friendly definition
Independent variable	The thing you change
Dependent variable	The thing you measure
Control variable	Something you keep the same
Continuous data	Number data that can have many possible values
Mean	The average value
Best-fit line	A line that shows the overall pattern of the results
Anomalous result	A result that does not fit the pattern

Equipment

Per group:

- Paper helicopter templates
- Scissors
- Paperclips
- Stopwatch
- Metre ruler
- Graph paper
- Exercise book or investigation sheet

Safety notes

- Students should not climb on chairs or benches to increase drop height.
- Use a sensible drop height that can be reached safely.
- Scissors should be used carefully.
- Students should stand clear of the drop zone.

Alternative investigation options

Students could investigate one of the following:

1. How does wing length affect fall time?
2. How does paperclip mass affect fall time?
3. How does drop height affect fall time?

Recommended option:

> *How does wing length affect the fall time of a paper helicopter?*

This is recommended because wing length is continuous and leads naturally to a line graph.

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Starter: Planning grid

Students complete this before starting the practical.

Question	Student answer
What is the independent variable?	
What is the dependent variable?	
What are three control variables?	
Is the independent variable continuous or categoric?	
Should you use a line graph or bar chart?	
Why?	

Suggested independent variable values

Students can test 5 different wing lengths, for example:

- 4 cm
- 5 cm
- 6 cm
- 7 cm
- 8 cm

Method

1. Make a paper helicopter with the chosen wing length.
2. Add the same number of paperclips each time.
3. Drop the helicopter from the same height.
4. Measure the time taken to reach the floor.
5. Repeat 3 times for the same wing length.
6. Calculate the mean fall time.
7. Repeat for the other wing lengths.
8. Plot a line graph of wing length against mean fall time.

Example results table

Students should ideally design this themselves. A support version is provided below.

Wing length / cm	Time trial 1 / s	Time trial 2 / s	Time trial 3 / s	Mean time / s
4				
5				
6				
7				
8				

Final assessed task

Students produce one page containing:

1. Investigation question.

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2. Variables.
3. Results table.
4. Mean values.
5. Correct graph.
6. Best-fit line.
7. Comment on anomalous results or errors.
8. Conclusion.

Success criteria

Skill	Success criteria
Variables	Independent, dependent, and control variables identified correctly
Table	Units, trials, and mean column included
Repeats	At least 3 trials for each wing length
Graph choice	Line graph chosen because wing length is continuous
Graph quality	Axes labelled, units included, sensible scale used
Best-fit line	Best-fit line drawn instead of dot-to-dot line
Anomalous results	Possible anomalous result identified or absence of anomalous results stated
Conclusion	Conclusion uses evidence from the results

Plenary

Students complete:

> *The most important thing I learned about drawing graphs is _____.*
> *The most important thing I learned about collecting data is _____.*
> *One thing I would improve next time is _____.*

Possible prep

Students write a short evaluation of their investigation, including:

1. One strength of their method.
2. One weakness of their method.
3. One possible random error.
4. One improvement they would make next time.

Using the workbook

The Quick check (slide 52) has its own box before the Exam corner. Students answer each question, then check it as you reveal the answer.

Teacher notes

This lesson can be used as a light-touch assessment of the whole unit. Students who are less confident may need a scaffolded results table, while more confident students can design their own table and choose their own wing lengths.

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